Question	Classification of Elements
Type	multiple choice
Option	${}^{1}_{1}H$ and ${}^{3}_{1}H$
Option	${}^{1}_{1}H$ and ${}^{4}_{2}He$
Option	${}^{1}_{1}H$ and ${}^{1}_{2}H$
Option	Both (a) and (c)
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following are the major drawbacks of the Proust hypothesis?
Type	multiple choice
Option	The position of hydrogen is not fixed
Option	After the discovery of fractional atomic mass, unable to explain it
Option	It failed after the discovery of metalloids
Option	All of the above
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Lavoisier's classification of elements is based on
Type	multiple choice
Option	Metal and Non-metal
Option	Metal and Metalloids
Option	Non-metal and Metalloids
Option	Both (b) and (c)
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The major drawbacks of Lavoisier's classification are:
Type	multiple choice
Option	He didn't give the idea of Nobel gas.
Option	He didn't explain the element having fractional atomic mass.
Option	He didn't introduce the idea of metalloids.
Option	All of the above
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following triads is a Dobereiner's Triad?
Type	multiple choice
Option	Li, Na, and K
Option	Ca, Sr, and Ba
Option	Cl, Br, and I
Option	All of the above
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Assertion: According to Dobereiner's a group of three similar elements were arranged so that the atomic weight of a middle element is equal to the arithmetic

	mean of the other two elements. Reason: Atomic weight is equal to 1/12th mass of a carbon atom.
Type	multiple choice
Option	Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
Option	Both Assertion and Reason are correct but Reason is not a correct explanation of Assertion.
Option	Assertion is correct but Reason is incorrect
Option	Assertion is incorrect but Reason is correct
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	In which set the elements Dobereiner's triads are not valid (i) Be (9), Mg (24), Ca (40) (ii) F (19), Cl (35.5), Br (80) (iii) C (12), N (14), O (16) (iv) S (32), Se (79), Te (128) (v)
Type	multiple choice
Option	Only (i)
Option	(i) and (ii)
Option	(ii), (iii), and (iv)
Option	(ii) and (iii)
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The major drawback of Doberiner's Triads are:
Type	multiple choice
Option	All elements do not follow the triads rule (like F, Cl, Br)
Option	He arranged elements only in groups (C, N, and O not valid)
Option	Only limited triads were able to be found by them
Option	All of the above
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Elements A, B, and C occur as Doberiner's triads. If the atomic mass of A is 40 and that of C is 137, what will be the atomic mass of B?
Type	multiple choice
Option	35
Option	120
Option	88
Option	74
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Newlands classification of elements did not include-
Type	multiple choice
Option	Metalloids
Option	Noble gases
Option	Metals

Option	Non-metals
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	What is Mendeleev's Periodic law?
Type	multiple choice
Option	Physical & chemical property of element are periodic function of their atomic weight.
Option	Property of element repeats after a certain value of atomic weight.
Option	Physical & Chemical property of elements are periodic function of their atomic number.
Option	Both (a) & (b)
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is not true for Mendeleev's Periodic table?
Type	multiple choice
Option	He arranged 63 elements in increasing order of atomic weight.
Option	He classified elements in 7 period & 9 groups.
Option	Group 1 to 8 was divided into two sub-groups.
Option	Sub-group (A) has normal element & sub-group (B) has transition element.
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Mendeleev's arranged _____ elements into _____ rows & _____ columns
Type	multiple choice
Option	65, 9, 7
Option	63, 7, 9
Option	63, 9, 7
Option	63, 7, 8
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following correct for Mendeleev's periodic table?
Type	multiple choice
Option	Group (1) to (7) was divided into two sub-group (A) & (B)
Option	Sub-group (A) & (B) contains s, P block & d-block respectively.
Option	Sub-group (A) & (B) contains main, normal element & Transition element respectively.
Option	All of these
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Group-8 in Mendeleev's periodic table contains (1) Set of three elements. (2) It contains total-9 elements. (3) It contains coinage metals. (4) It contains Nobel gas.
Type	multiple choice

Option	1, 2 & 3
Option	1 & 2
Option	1, 3 & 4
Option	All of these
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The correct statements regarding s-block are: 1) Last electron enters into outermost ns subshell. 2) All inner shell upto (n-1) are complete filled. 3) s-orbital has two electron & hence two group in periodic table. 4) s-block has group 13 to 18 element. 5) General electronic configuration is [I.G.] ns^{1-2} & it shows +1 and +2 oxidation state.
Type	multiple choice
Option	1 & 2 only
Option	1, 3, 4 & 5 only
Option	1,2,3 & 5 only
Option	All of these
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The correct statement for p-block element is: 1) Last electron enters into outermost np-subshell. 2) G.E.C. – [IG] $ns^2\ np^{1-6}$ 3) It include metal, non-metal & metalloid, shows variable valency. 4) It consist solid, liquid & gas. 5) All inner shell up-to (n-1) are completely filled.
Type	multiple choice
Option	1 & 2 only
Option	1, 3, 4 & 5 only
Option	1, 2, 3 & 4 only
Option	All of these
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is not true regarding d-block element-
Type	multiple choice
Option	Last electron enters into penultimate d-subshell
Option	d-Block element lies in group 3 to 12 & G.E.C. - [I.G.] $(n-1)d^{1-10}ns^{1-2}$
Option	All inner shell up-to (n-2) are completely filled.
Option	All d-block elements are Transition element.
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	All d-block elements are transition element except:
Type	multiple choice
Option	Zn, Cd, Hg
Option	Cu, Au, Ag

Option	Cr, Mn, W
Option	Pt, Pd, Ag
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is not true for Transition Element?
Type	multiple choice
Option	All Transition elements are called d-block elements.
Option	They have at least one unpaired delectron in ground state or in their common oxidation state.
Option	Cu is a transition element.
Option	Pd has electronic configuration $4d^9\ 5s^1$.
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is not true for fblock element?
Type	multiple choice
Option	Last electron enters into $(n-2)f-$subshell
Option	G.E.C. is $(n-2)\ f^{1-14}(n-1)d^{0,1,2}\ ns^2$
Option	It is also known as inner Transition Element.
Option	It contains 4f and 5f series as Lanthanide & Actionoid respectively.
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Match different type of element in column-I with their position in periodic table column-II:

SN	Column-I (different type of element)		Column-II (position in Periodic table)
1	Lanthanoid	(A)	58-71
2	Actinoid	(B)	90-103
3	Transition Element	(C)	Ce-Lu
		(D)	Th-Lr
		(E)	6th Period
		(F)	7th Period
		(G)	4f series
		(H)	5f series
		(I)	G-3 to G-11
		(J)	d-Block
		(K)	f-block

Type	multiple choice
Option	1- (A), (C), (E), (K),(G) 2-(B), (D), (F), (H),(K) 3-(E),(F), (1), (J)
Option	1-(J), (C), (E), (K), 2-(B), (D), (F), (H) 3-(F), (1), (A)
Option	1- (A), (1), (E), (K), 2-(B), (D), (F), (H) 3-(F), (C), (J)
Option	1-(A), (C), (F), (K), 2-(B), (D), (F), (H) 3-(E), (1), (J)
Answer	1
Solution	

Positive Marks	4
Negative Marks	1
Question	Match different type of period in column-I with their position in column II:

S.N.	Column-I (Type of period)		Column-II (Position in P.T)
1	Very short period	(A)	1
2	Short period	(B)	2 & 3
3	Long period	(C)	4 & 5
4	Longest period	(D)	6 & 7

Type	multiple choice
Option	1- (B), 2-(C), 3-(A), 4-(D)
Option	1- (D), 2-(B), 3-(C), 4-(A)
Option	1- (A), 2-(B), 3-(C), 4-(D)
Option	1- (B), 2-(A), 3-(C), 4-(D)
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Match different type of period in column-I with number of elements in column-II:

S.N.	Column-I (period)		Column-II (Number of elements)
1	Very short period	(A)	2
2	Short period	(B)	8
3	Long period	(C)	18
4	Longest period	(D)	32

Type	multiple choice
Option	1- (B), 2-(C), 3-(A), 4-(D)
Option	1 (A), 2-(B), 3-(C), 4-(D)
Option	1- (D), 2-(C), 3-(A), 4-(B)
Option	1- (B), 2-(A), 3-(C), 4-(D)
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Maximum number of elements in 8th period will be:
Type	multiple choice
Option	32
Option	18
Option	8

Option	50
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is a merits of Mendeleev's periodic table? 1) Element with similar properties was placed in similar group. 2) He predicted some element which were discovered later. 3) Ga, Sc were known as eka Aluminum & Eka Boron. 4) Eka silicon & Eka manganese were later known as Ge & Te respectively.
Type	multiple choice
Option	1 & 2 only
Option	1, 2 & 3
Option	1, 2, 3& 4
Option	3 & 4 only
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Mendeleev's corrected atomic weight of: (1) U, Be (2) Pt, Au, In (3) In, Zn, Sn (4) Cu, Mn, Na
Type	multiple choice
Option	1 & 2 only
Option	1, 2 & 3
Option	1, 2, 3 & 4
Option	3 & 4 only
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Match Mendeleev's elements in column I with present element in column II:

S.N.	Column-I (Mendeleev's Element)		Column-II (Present Element)
1	Eka-Al	(A)	Ga
2	Eka-Bo	(B)	Sc
3	Eka-Si	(C)	Ge
4	Eka-Mn	(D)	Te

Type	multiple choice
Option	1 - (B), 2-(C), 3-(A), 4-(D)
Option	1- (D), 2-(B), 3-(C), 4-(A)
Option	1- (A), 2-(B), 3-(C), 4-(D)
Option	1- (B), 2-(A), 3-(C), 4-(D)
Answer	3
Solution	
Positive Marks	4
Negative Marks	1

Question	Which of the following is not true for demerits of Mendeleev's periodic table?
Type	multiple choice
Option	Position of Hydrogen is ambiguous.
Option	Elements with similar property were placed in different group and element with different property were placed in same group.
Option	No separate position for isotopes.
Option	Heavy elements were placed after lighter Element in Mendeleev's periodic table.
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Mendeleev's violets his periodic law because of:
Type	multiple choice
Option	Ar, K
Option	Co, Ni
Option	Te, I
Option	All of these
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is correct for main element? 1) s & p-block. 2) Also known as normal & representative element 3) nth shell or outermost shell is incomplete. 4) s & p-block except inert gas.
Type	multiple choice
Option	1, 2 & 3 only
Option	2, 3 & 4 only
Option	1, 2, 3 & 4
Option	1 & 2 only
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The element of 3rd period which shows property of entire group is known as:
Type	multiple choice
Option	Normal element
Option	Representative element
Option	Main Element
Option	Typical Element
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Assertion: 2nd period element is not considered as Typical element. Reason: 2nd period element has small size high I.E. & high E.N. & absence of d-orbital.
Type	multiple choice
Option	Both assertion and reason are correct and reason are correct explanation of assertion.
Option	Both assertion and reason are correct and reason is not correct explanation of assertion.
Option	Assertion is true but reason is false.

Option	Both assertion and reason are false.
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is not true for diagonal relationship?
Type	multiple choice
Option	Element having same ionic potential shows diagonal relationship.
Option	(Li & Mg), (Be, Al); (B, Si) shows diagonal relationship.
Option	Element having same charge & size ratio shows diagonal relationship.
Option	Element with 3rd period & 4th period shows diagonal relationship.
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Match different type of element in column-I with their position in periodic table column-II:

Column-I (Type of Element)		**Column-II (position in Periodic table)**	
1	Transition Element	(A)	Z > 100
2	Inner transition element	(B)	F-block
3	Trans Uranic element	(C)	Z > 92
4	Transferming element	(D)	G-3 to G-12

Type	multiple choice
Option	1- (B), 2-(C), 3-(A), 4-(D)
Option	1- (D), 2-(B), 3-(C), 4-(A)
Option	1- (D), 2-(B), 3-(C), 4-(A)
Option	1- (B), 2-(A), 3-(C), 4-(D)
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The electronic configuration of an element is $1s^2 2s^2 3p^6 3s^2 3p^3$. What is the atomic number of the element, which is just below the above element in the Periodic table?
Type	multiple choice
Option	36
Option	49
Option	33
Option	34
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	$4d^3 3s^2$ configuration belongs to which group
Type	multiple choice
Option	II A
Option	II B
Option	V B

Option	III B
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	If an atom has electronic configuration $3s^2 2s^2 2p^6 3s^2 3p^6 3d^3 4s^2$, it will be placed in
Type	multiple choice
Option	Second group
Option	Third group
Option	Fifth group
Option	Sixth group
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The elements having the electronic configuration $[Kr]4d^{10}, 4f^{14}, 5s^2, 5p^6 5d^2, 6s^2$ belong to
Type	multiple choice
Option	s-block
Option	p-block
Option	d-block
Option	f-block
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	An element has electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^4$. Predict their period, group and block
Type	multiple choice
Option	Period = 3rd, block = p, group = 16
Option	Period = 5th, block = s group = 1
Option	Period = 3rd, block = p, group = 10
Option	Period = 4th, block = d, group = 12
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	In an Atom, what does Z represent?
Type	multiple choice
Option	Number of protons
Option	Atomic number
Option	Number of neutrons
Option	Both (1) and (2)
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The correct statement about effective nuclear charge is:
Type	multiple choice
Option	Attraction between nucleus & outermost electron.
Option	Attraction between nucleus & outermost electron despite repulsion of inner

	electron.
Option	The attraction between nucleus & inner electron
Option	The repulsion between inner electron & outer electron.
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Z-effective is valid for:
Type	multiple choice
Option	Single e^- system
Option	Single or multiple e^- system
Option	Multi e^- system
Option	None of these
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is true for Z-effective?
Type	multiple choice
Option	It is net force of attraction between nucleus & outer e^-
Option	Z-effective is $f_{net} = f_{att} - f_{rep}$
Option	Z-effect is $f_{net} = f_{rep} - f_{att}$
Option	Both (1) and (2)
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The correct definition about for shielding effect is:
Type	multiple choice
Option	repulsion between inner electron and outermost electron.
Option	attraction between inner electron and outermost electron.
Option	attraction between nucleus and outermost electron.
Option	All of these
Answer	1
Solution	
Positive Marks	4
Negative Marks	1